

TASK 1 – Data Collection and Dataset Understanding

Aim:

To understand the dataset and identify the columns, data types, and size of the dataset.

Code:

```
import pandas as pd

data = pd.read_csv("StudentsPerformance.csv")

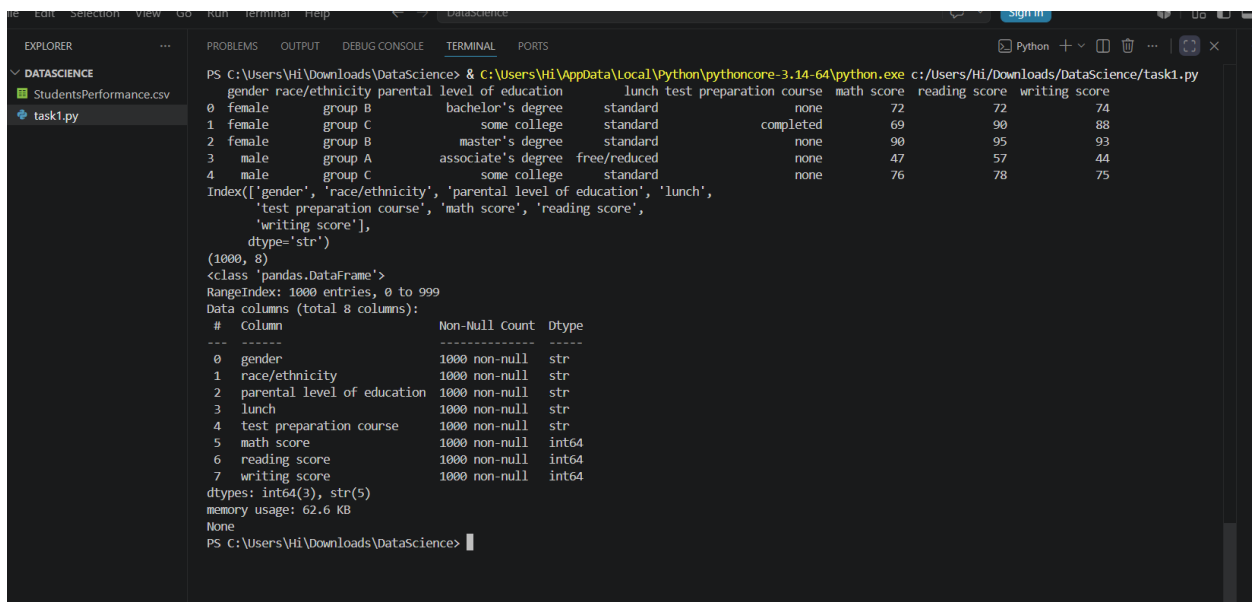
print(data.head())

print(data.columns)

print(data.shape)

print(data.info())
```

Output:



```
PS C:\Users\Hi\Downloads\DataScience> & c:\Users\Hi\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/Hi/Downloads/DataScience/task1.py
gender race/ethnicity parental level of education lunch test preparation course math score reading score writing score
0 female group B bachelor's degree standard none 72 72 74
1 female group C some college standard completed 69 90 88
2 female group B master's degree standard none 90 95 93
3 male group A associate's degree free/reduced none 47 57 44
4 male group C some college standard none 76 78 75
Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
      'test preparation course', 'math score', 'reading score',
      'writing score'],
      dtype='str')
(1000, 8)
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
# Column Non-Null Count Dtype
---
0 gender 1000 non-null str
1 race/ethnicity 1000 non-null str
2 parental level of education 1000 non-null str
3 lunch 1000 non-null str
4 test preparation course 1000 non-null str
5 math score 1000 non-null int64
6 reading score 1000 non-null int64
7 writing score 1000 non-null int64
dtypes: int64(3), str(5)
memory usage: 62.6 KB
None
PS C:\Users\Hi\Downloads\DataScience>
```

Description:

This dataset contains student performance details such as gender, parental education, lunch type, and scores. The dataset has 1000 rows and 8 columns.

Conclusion:

The dataset was loaded successfully and the structure of the dataset was understood clearly.